Grading Rubrics

CSCI 416/516 Final Exam

Name:

Student ID:

Before you start: Check your exam. The exam has 5 pages and ~~13~~ 12 questions in total. If your exam is not printed clearly or incomplete, let the instructors know and we will give you a new copy of the exam.

Submission: Write down your name and student ID. You have 3 hours to complete your exam. You are allowed a one-sided (US letter-sized) cheatsheet and a basic or scientific calculator. For all the questions except the binary choice questions, **please show your work/process on how you reach the conclusions to receive full credits assigned to the questions.**

+1 B • **Problem 1 [1 pt(s)]: Euclidean Distance.**
In high-dimensional spaces, does the Euclidean distance metric become more effective at distinguishing between different data points, in the context of KNN?

- (A) Yes
- (B) No

+1 A • **Problem 2 [1 pt(s)]: Gradient Descent.**
Is Stochastic Gradient Descent (SGD) more computationally efficient per iteration than Batch Gradient Descent?

- (A) Yes
- (B) No

+1 A • **Problem 3 [1 pt(s)]: Linear Regression.**
What is the linearity assumption in linear regression?

- (A) The linear relationship between features and the predicted variable
- (B) The linear relationship between features themselves

+1 A • **Problem 4 [1 pt(s)]: Support Vector Machine.**
Why do we want to use kernel tricks in SVM?

- (A) Kernel tricks allow the efficient performance of a non-linear classification without explicitly transforming the data
- (B) Kernel tricks mitigate the effect of the curse of dimensionality in high-dimensional spaces

- **Problem 5 [2 pt(s)]: 1D Convolution.**
Given a kernel $k = \{2, 9, 3\}$ and a vector $v = \{1, 2, 3\}$, what is the result of $k * v$? After getting the result $k * v$, what is the new result after applying ReLU on $k * v$? Show the work that lead to your conclusion.

$$k * v = \boxed{\begin{aligned} &2 \times [1,2,3,0,0] \\ &+ 9 \times [0,1,2,3,0] \\ &+ 3 \times [0,0,1,2,3] \end{aligned}} = [2,4,6,0,0] = [2,13,27,33,9]$$

$$\begin{aligned} &+ [0,9,18,27,0] \\ &+ [0,0,3,6,9] \end{aligned}$$

+1

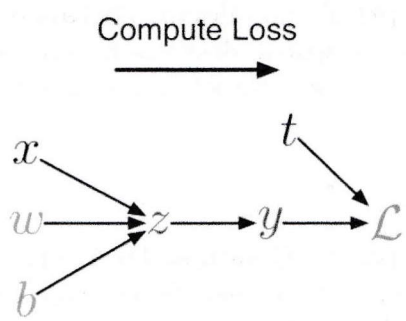
$$\text{ReLU}(k*v) = \boxed{[2,13,27,33,9]} \; +1$$

Award +1 (partial credits) if $k*v$ was wrong but ReLU was correctly applied.

- **Problem 6 [3 pt(s)]: Backpropagation.** What does the back pass look like (in terms of the error signals of parameters/activations, such as but not limited to b, w), given the illustrated forward pass in Figure below? Show your work on how the conclusion is reached.

Computing the loss:

$$z = wx + b$$
$$y = \sigma(z)$$
$$\mathcal{L} = \frac{1}{2}(y - t)^2$$



$$\bar{\mathcal{L}} = \frac{\partial \mathcal{L}}{\partial \mathcal{L}} = \boxed{1} \; +0.5$$

+0.5 for not doing $\bar{x}, \bar{t}$.

$$\bar{y} = \boxed{\bar{\mathcal{L}} \cdot \frac{\partial \mathcal{L}}{\partial y}} \; +0.5 = \bar{\mathcal{L}} \cdot (y - t) = (y - t)$$

$$\bar{z} = \boxed{\bar{y} \cdot \frac{\partial y}{\partial z}} \; +0.5 = \bar{y} \cdot \sigma'(z) = (y - t) \cdot \sigma'(z)$$

$$\bar{w} = \boxed{\bar{z} \cdot \frac{\partial z}{\partial w}} \; +0.5 = \bar{z} \cdot x = (y - t)\, \sigma'(z) \cdot x$$

$$\bar{b} = \boxed{\bar{z} \cdot \frac{\partial z}{\partial b}} \; +0.5 = \bar{z} = (y - t) \cdot \sigma'(z)$$

	Cloudy	Not Cloudy
Raining	50/100	2/100
Not Raining	24/100	24/100

- **Problem 7 [2 pt(s)]: Joint Entropy.**
  Suppose $X = \{\text{Raining, Not raining}\}$, $Y = \{\text{Cloudy, Not Cloudy}\}$, give the above Table. What is the joint entropy, $H(X, Y)$? Show your work on how the conclusion is reached.

$$H(X,Y) = -\sum_{x \in X} \sum_{y \in Y} P(x,y) \log_2 P(x,y)$$

$$= \boxed{-\left[\frac{50}{100}\log_2\frac{50}{100} + \frac{2}{100}\log_2\frac{2}{100} + \frac{24}{100}\log_2\frac{24}{100} + \frac{24}{100}\log_2\frac{24}{100}\right]}$$

+2

$$\approx 1.60 \text{ bits}$$

- **Problem 8 [2 pt(s)]: Specific Conditional Entropy.**
  Following the setup from the previous question, what is the entropy of cloudiness Y, given that it is raining (X = Raining)? Show your work on how the conclusion is reached.

$$H(Y \mid X = \text{raining}) = -\sum_{y \in Y} P(y \mid x = \text{raining}) \log_2 P(y \mid x = \text{raining})$$

$$P(\text{cloudy} \mid \text{raining}) = \frac{P(\text{cloudy, rain})}{P(\text{rain})} = \frac{50}{100} \times \frac{100}{52} = \frac{50}{52}$$

$$P(\neg\text{cloudy} \mid \text{rain}) = \frac{P(\neg\text{cloudy, rain})}{P(\text{rain})} = \frac{2}{100} \times \frac{100}{52} = \frac{2}{52}$$

$$H(Y \mid x = \text{rain}) = \boxed{-\left[\frac{50}{52}\log_2\frac{50}{52} + \frac{2}{52}\log_2\frac{2}{52}\right]} \approx 0.24 \text{ bits}$$

+2

- **Problem 10 [2 pt(s)]: AdaBoost.**
Explain what this formula does, in the context of AdaBoost by answering: (1) what is this formula? (2) what does the part $y_i \log h_{\boldsymbol{\theta}}(\boldsymbol{x}_i) + (1 - y_i)\log(1 - h_{\boldsymbol{\theta}}(\boldsymbol{x}_i))$ do? (3) what does the part $\lambda ||\boldsymbol{\theta}_{[1:d]}||_2^2$ do? (4) What is the term $w_{i,t}$ and how does it affect $\mathcal{J}_{\text{reg},t}(\boldsymbol{\theta})$?

$$\mathcal{J}_{\text{reg},t}(\boldsymbol{\theta}) = -\sum_{i=1}^{n} w_{i,t}[y_i \log h_{\boldsymbol{\theta}}(\boldsymbol{x}_i) + (1 - y_i)\log(1 - h_{\boldsymbol{\theta}}(\boldsymbol{x}_i))] + \lambda ||\boldsymbol{\theta}_{[1:d]}||_2^2 \qquad (1)$$

+0.5

(1) the $\boxed{\text{cost (regularized) function}}$ for the base classifier/learner at the tth iteration

+0.5

(2) the $\boxed{\text{loss}}$, given the ith training sample.

+0.5

(3) $\boxed{\text{regularize}}$ the parameter matrix

+0.5

(4) $W_{i,t}$ = the $\boxed{\text{weight}}$ of the ith training sample for/at the tth iteration.

- **Problem 11 [3 pts]: Support Vector Machines.**
The optimization objective of SVM is given as $\min_{\boldsymbol{\theta}} C \sum_{i=1}^{N} [y_i \text{cost}_1(\boldsymbol{\theta}^\top \boldsymbol{x}_i) + (1 - y_i)\text{cost}_0(\boldsymbol{\theta}^\top \boldsymbol{x}_i)] + \frac{1}{2}\sum_{j=1}^{d} \theta_j^2$, where cost_0 and cost_1 are defined using the hinge loss. Explain the difference between the scenario in which the tunable hyperparameter C is large and the scenario in which C is small - what are we favoring, by making C large or small?

Large C: favoring minimizing the classification error over maximizing the margin. +1.5

Small C: favoring maximizing the margin over minimizing the classification error +1.5

- **Problem 12 [2 pts]: Lagrangian Multipliers.**
  Suppose a sample in the training dataset has a Lagrangian multiplier being 0. What does this say about this sample?

this sample is not a support vector. +2

award full credits if the student describes what a non SV does (or doesn't) do, even without explicitly stating that the sample is not a SV.

- **Problem 13 [3 pts] (bonus): Attention & Transformers.**
  What's the relationship between the Scaled Dot-Product Attention and Multi-Head Attention, in the transformer architecture?

The Multi-Head Attention (block) utilizes multiple Scaled Dot-Product Attention. +3